Minimum Wage

Introduction

Does raising the minimum wage genuinely benefit businesses and their workers, or does it have unanticipated consequences for the same communities it is intended to support? Raising the minimum wage is commonly justified as a way to improve workers' quality of life and reduce poverty, but the impact on family-owned and locally operated businesses presents a more worrisome picture. These smaller companies, in contrast to large corporations, sometimes lack the financial flexibility to accommodate rising labor costs, forcing them to make difficult decisions like staff reductions, price increases, or full closure, revealing the downsides of a rise in minimum wage.

This essay examines how a minimum wage increase affects employment levels, pricing strategies, and business sustainability by utilizing the data from 410 fast-food establishments across New Jersey and Pennsylvania. The data indicates that while large corporations may be able to withstand these changes, local businesses face significant challenges that could endanger their longevity and contributions to the economy. Given this evidence, this essay argues that raising the minimum wage harms small businesses more than it helps, and aims to demonstrate the importance of modifying wage policies to protect the needs of family-owned businesses.

Theory and Hypotheses

The debate over raising the minimum wage revolves around whether or not it will benefit businesses and employees. On the one hand, it is argued that higher wages increase

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workers' purchasing power, which boosts consumer spending and drives demand for goods and services. As a result, businesses may see increased revenues, which can help offset rising labor costs. Wage increases, on the other hand, place significant financial burdens on small businesses, particularly those owned by families. Unlike larger corporations, these businesses frequently have lower profit margins and lack the resources to adapt to rising labor costs. As a result, they may respond by cutting jobs, reducing employee hours, raising prices, or even shutting down.

Previous research has yielded conflicting results on this topic. David Card and Alan Krueger's study of fast-food restaurants in New Jersey and Pennsylvania discovered no significant reductions in employment following a minimum-wage increase, implying that businesses largely adapted to the changes. However, further investigation reveals significant differences among various types of businesses. Larger franchises, with economies of scale and more financial flexibility, were better able to withstand rising costs. In contrast, family-owned businesses faced more difficult challenges, such as closures and fewer job opportunities. These findings demonstrate that the impact of minimum wage increases varies significantly by business size and structure, with family-owned businesses bearing a disproportionate burden.

This produces two central hypotheses. The null hypothesis asserts that raising the minimum wage has no significant impact on employment or business closures in the fast-food industry. The alternative hypothesis contends that raising the minimum wage decreases employment and increases the likelihood of business closures, particularly among family-owned businesses. These hypotheses are tested using a regression model in which employment level is

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the dependent variable, minimum wage is the primary explanatory variable, and meal prices and ownership type serve as controls. The model can be expressed as:

$$Employment\ Level = \beta_0 + \beta_1 + \beta_2 + \beta_3 + \varepsilon$$

In this model, β_1 represents the effect of the minimum wage on employment, while β_2 and β_3 represent the influence of other variables. The significance level for hypothesis testing is set at $\alpha = 0.05$. This study aims to determine the degree to which minimum wage increases affect small businesses and to provide insight into policies that might minimize negative outcomes. By focusing on family-owned businesses, this model highlights the broader economic implications of wage policy changes.

Data

This analysis uses data from 410 fast-food restaurants in New Jersey (NJ) and Pennsylvania (PA) to assess the impact of raising the minimum wage on employment and business outcomes. The dataset includes key variables like wages (measured in dollars per hour), employment levels (measured as total number of employees), meal prices (measured as average meal price), and ownership type (company-owned or franchise). This dataset is especially useful for comparing outcomes in New Jersey, which raised the minimum wage, and Pennsylvania, which did not, because it provides a quasi-experimental framework for the analysis.

The study's dependent variables are employment and business closures, which show how minimum wage increases affect the labour market and business viability. The minimum wage is used as the primary explanatory variable, with controls for meal price and ownership type to account for any potential confounding factors. Meal prices demonstrate how businesses

adjust their pricing strategies in response to rising labor costs, whereas ownership type distinguishes corporate from family-owned businesses. Missing data for key variables was removed using listwise deletion, producing a consistent sample size and simplified analyses.

Table 1 presents descriptive statistics for the key variables, such as mean and median values and standard deviation. This table from the "Minimum Wage: Data Description" PDF shows that wages in New Jersey have risen to an average of $5.08, compared to $4.62 in Pennsylvania. NJ had an average of 21.03 employees, slightly more than PA's 21.17, and meal prices increased slightly more in NJ ($3.41) than in Pennsylvania ($3.03).

Variables	Mean	Median	Standard Deviation
Wages (NJ)	5.08	5.05	0.10
Wages (PA)	4.62	4.50	0.36
Employment (NJ)	21.03	20.50	9.29
Employment (PA)	21.17	20.00	8.28
Prices (NJ)	3.41	3.17	0.64
Prices (PA)	3.03	2.84	0.57

Table 1

The visual representation of data trends is critical for understanding the effects of the minimum wage increase. Figure 1, a bar chart comparing employment levels before and after the wage increase, and Figure 2, a line graph depicting meal price changes over time, are critical components of the analysis. These plots show that, while employment in New Jersey increased slightly following the wage hike, it decreased slightly in Pennsylvania. Similarly, meal prices increased modestly in New Jersey but remained stable in Pennsylvania.

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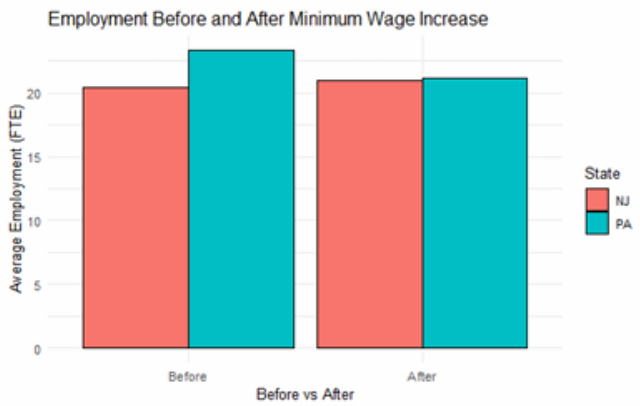

Figure 1

Figure 2

A scatterplot of employment versus wages tells you more about the relationship between the two variables. This plot shows a weak positive correlation, implying that wages are not a good predictor of employment levels. These visuals, along with descriptive statistics, serve as the foundation for examining the impact of minimum wage increases on family-owned businesses.

Figure 3

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While overall employment levels did not fall significantly, the burden of higher labor costs was most likely borne disproportionately by small, family-owned businesses, as shown by the descriptive statistics and plots. These findings provide the foundation for more in-depth statistical analyses, such as regression modeling, to investigate these effects further.

Regression Models

This study uses regression modeling to investigate the relationship between minimum wage increases and employment levels, focusing on both bivariate and multivariate approaches. The bivariate regression looks at the direct relationship between employment rates (dependent variable) and the minimum wage (explanatory variable). To isolate the effects of the minimum wage, the multivariate regression uses control variables such as meal price and ownership type. This method helps to account for potential confounders that may influence employment outcomes. The bivariate regression model is specified as:

$$Employment\ Level\ =\ \beta_0 + \beta_1 +\ \epsilon$$

In this model, β_1 denotes the estimated impact of the minimum wage on employment. The study found a weak positive association, with a slope of 0.44 and a p-value of 0.803. This implies that the minimum wage has no discernible effect on employment levels in this case. To better understand the dynamics, a multivariate regression model was employed:

$$Employment\ Level\ =\ \beta_0 + \beta_1 + \beta_2 + \beta_3 +\ \epsilon$$

Here, β_2 and β_3 determine the meal price and ownership type. Table 2 shows that, while the coefficient for minimum wage (2.39, $p = 0.165$) remains statistically insignificant, meal price (-5.05, $p < 0.001$) and ownership type (-2.13, $p = 0.01962$) have significant impacts on

employment levels. These findings suggest that pricing strategies and business structure have a greater impact on employment outcomes than minimum wage increases alone.

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Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  26.9958     8.5161   3.170  0.00166 **
wage_st2      2.3914     1.7208   1.390  0.16548
pmeal2       -5.0497     0.6747  -7.484 5.62e-13 ***
co_owned     -2.1258     0.9069  -2.344  0.01962 *
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Table 2

The regression analysis includes diagnostic metrics to assess model performance. The multivariate model's R-squared value of 0.156 indicates that it explains 15.6% of the variation in employment rates. This is an improvement over the bivariate model, which had an adjusted R-squared value near zero. However, it emphasizes that other factors that the model does not account for are likely to influence employment outcomes.

Figure 4 shows a plot of predicted and actual employment levels, demonstrating the model's performance. Even after controlling for other variables, the relationship between minimum wage and employment remains weak. The residuals are evenly distributed across the predicted line. Figure 5 shows that the model has no significant heteroskedasticity, indicating consistent variance across all fitted values.

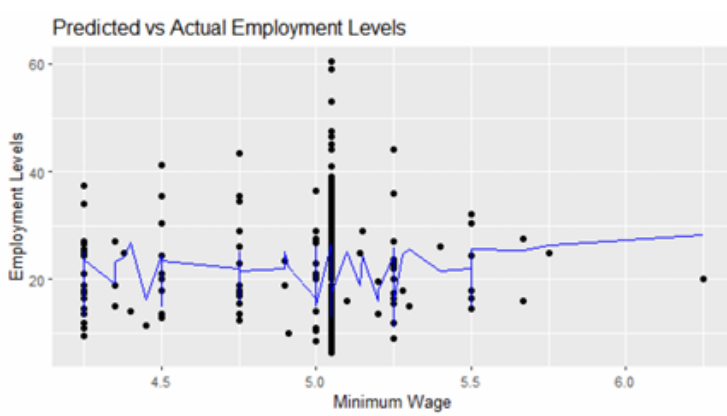

Figure 4

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The overall regression results show that raising the minimum wage has no significant direct impact on employment in the fast-food industry. The significant negative coefficients for meal price and ownership type, however, indicate that smaller, family-owned businesses will have a more difficult time adjusting to wage increases. These findings support the notion that minimum wage increases disproportionately affect small businesses and demonstrate the policy's uneven effects. The findings highlight the importance of enacting specific legislation to mitigate these effects and help family-owned businesses maintain profitability and job opportunities.

Model Diagnostics

Model diagnostics are critical to ensure the validity and reliability of the regression analysis. This section evaluates the assumptions of ordinary least squares (OLS) regression, examines the model's performance, and identifies potential areas for improvement.

The first diagnostic test involves examining residual plots to assess heteroskedasticity and non-linearity. The residual plot shows a random scatter of residuals around zero without a discernible pattern. This indicates that the variance of errors is consistent across fitted values, suggesting no significant evidence of heteroskedasticity. Additionally, a formal non-constant variance test produces a p-value of 0.43969, confirming that the assumption of constant variance holds.

Next, the Q-Q plot, figure 6, is used to assess the normality of residuals. The plot reveals minor deviations from normality, particularly in the tails, which suggests that the residuals are not perfectly normally distributed. This finding is further supported by the Shapiro-Wilk normality test, which yields a statistic of 0.9232 and a p-value of less than 0.001, indicating a

significant deviation from normality. While the violation of the normality assumption may slightly impact the reliability of p-values and confidence intervals, the overall regression results remain interpretable given the robustness of OLS to minor deviations from normality.

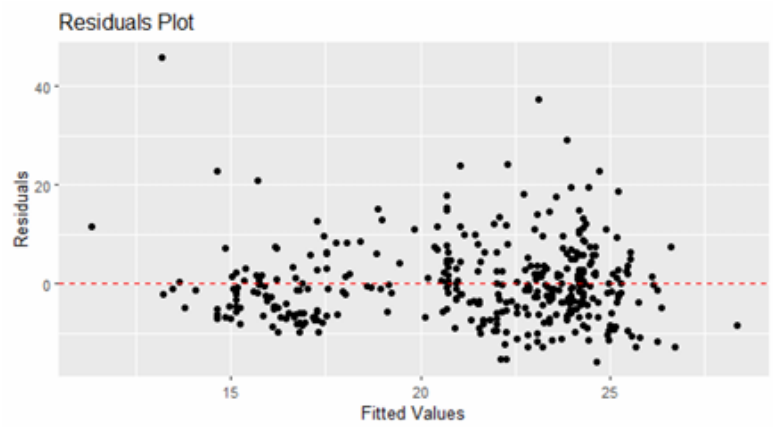

Figure 5

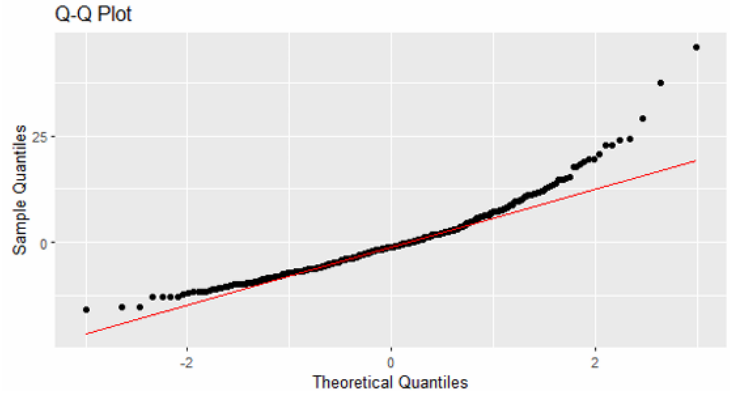

Figure 6

Outlier analysis is another crucial diagnostic. The Bonferroni outlier test identifies two significant outliers in the dataset. These outliers, while potentially influential, were retained in the analysis as they represent valid observations within the sample and reflect real-world variability in business outcomes. However, their presence suggests caution when interpreting the results, as they may slightly distort the estimated relationships.

Finally, the model's explanatory power is evaluated using R-squared and adjusted R-squared values. The multivariate regression model achieves an R-squared of 0.156 and an

adjusted R-squared of 0.1489, indicating that the model explains approximately 15.6% of the variability in employment levels. While this is a modest level of explanatory power, it suggests that additional factors beyond minimum wage, meal price, and ownership type are likely influencing employment outcomes. The relatively low R-squared also points to the need for further research to identify other relevant variables or employ alternative modeling approaches.

In summary, the diagnostic tests reveal that the multivariate model generally meets OLS assumptions, with the exception of normality, which shows minor deviations. There is no evidence of heteroskedasticity or non-linearity, and while outliers are present, they do not substantially compromise the model's validity. The modest R-squared value highlights the complexity of employment dynamics and suggests that minimum wage increases, while not statistically significant in this analysis, may interact with other unmeasured factors. These diagnostics provide confidence in the robustness of the results, while also underscoring the limitations and opportunities for further refinement.

Discussion

While minimum wage rises have no statistically significant direct influence on overall employment numbers, they still present considerable obstacles to family-owned and local companies. Larger organizations can handle higher payments, while smaller businesses are more likely to close or reduce employment, according to regression models and prior research. The research question, whether raising the minimum wage benefits or harms firms, gives a clear answer: while the policy may not have a major impact on overall employment, it

disproportionately harms small enterprises. Because of lower profit margins and less resources, these businesses struggle more than larger companies.

Because this is an observational study, causation cannot be established definitively. While regression models demonstrate consistent patterns, other unmeasured variables may impact the outcome. Future research that uses approaches such as difference-in-differences analysis may provide stronger evidence of causal relationships. The findings have significant policy consequences. While raising the minimum wage intends to alleviate poverty and enhance worker well-being, it may unintentionally affect small firms, resulting in closures and fewer job openings. Policymakers should adopt complementary measures, such as tax incentives or subsidies, to help small businesses adjust.

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